

MULTIMEDIA

Describe Multimedia Element		Explain the purpose of each multimedia elements
Text	Combination of letters that form words and use to convey text information to users	1. Text is used as a field for reading to convey information 2. Text is used as a caption to emphasize information
Graphic	Digital representation of non-text information such as image, photograph, chart and drawing.	1. Graphic can be used to emphasize information more clearly than just using text. 2. Graphic can make presentation more exciting when used as background image or main focus object.
Audio	Produced by vibration as perceive by sense of hearing in a form of speech, sound effect and music.	1. Audio can add exciting dimension to a presentation when used as background music or sound effect 2. Audio narration can increase user's understanding because it associate sound with information.
Animation	Series of graphical images displayed one after another to stimulate movement.	1. Animation can help explaining technical concept more clearly 2. Animation can be used to illustrate the movement of things that change over time
Video	Series of frame containing picture and sound played back at fast rate.	1. Video can evoke user's emotion by bringing elements that resemble actual situation 2. Video can convey realistic information and help to retain more information in memory

Differentiate between Multimedia Authoring Tools:

Types	Characteristic, usage and example	Usage	Example
Time-based	Multimedia elements and events are organized along a timeline.	suitable for creating animation and video	Adobe Animate, Adobe Flash
Page-based	Multimedia elements and events are organized as pages in a book or stack of cards.	Suitable for presentation of education content	Canva, PowerPoint
Icon-based	Multimedia elements and events are organized in a structural framework as an object in flowchart	Suitable for interactivity	Scratch, Adobe Authorware

Explain Multimedia navigation:

Hypertext	Hypermedia	Tab	Audio	Hand gesture	Menu
related to link plain text to other text based document	connecting hypertext with other media such as graphics, sounds, animation and video	the clickable area at the top that shows another page or area.	control and handling multimedia applications using audio.	control multimedia applications through hand gesture recognition scheme.	user navigates by pressing a key, clicking a mouse, or pressing a touch screen

Differentiate offline and online Multimedia delivery:

Criteria	Acessibility	Medium
Online	Use internet to deliver the content	Web-based, cloud
Offline	Do not need to use internet to deliver the content	Hard disk, cd/dvd ROM, SSD

Differentiate 2D, 3D, 4D		
2D	3D	4D
Represent in 2 dimension: length and width	Represent in 3 dimension: length, width and depth	Represent in 4 dimension: length, width, depth, time and motion
Use x-axis and y-axis	Use x-axis, y-axis and z-axis	Use x-axis, y-axis and z-axis to a space with 4 dimension
Flat shape: circle, square	Life-like shape: sphere, cube	Abstract shape: polychora, polyhedral

Describe pre-production, production and post production phase

Pre-production	The process of planning and generating ideas to develop the multimedia product Process involves: 1. Planning 2. Resource organization 3. Flowchart 4. Navigation 5. Define action screen
Production	The process of producing multimedia product by using the authoring tools Process involves: 1. Digitizing arts 2. Modelling and animation 3. Shooting video 4. Recording audio 5. Authoring 6. Proof reading 7. Quality control
Post production	The process of assembling, editing and testing the final product Process involves: 1. Testing 2. Mastering 3. Archiving 4. Duplication 5. Distribution

DATABASE

Differentiate between File Processing System and Database System:	
File Processing System	Database System
Data in a file based system is stored in separate files in different department	The data in a database environment is shared, usually over a network, by the entire organization
Less secure because file processing system usually do not protected by password	More secure because database system is protected by username and password
Data can be redundant because data may be duplicate in different files	Less data redundancy because database system do not allowed data duplication
Design for the use of single user	Design for the use of multiple users

Identify advantages and disadvantages of using Database System:	
Advantages	Disadvantages
Higher data integrity because data in a database are consistent and accurate	Database systems are often complex, so the training for users to use the system is required.
Data is secure because only authorized user can access the database system	The cost of developing and maintaining the database system is high.
The data in a database environment is shared, usually over a network to be accessed by the entire organization.	the use of the same program at a time by many users sometimes lead to the loss of some data.

Explain the concept of data integrity:	
Database system has high data integrity because data in a database system are consistent and accurate.	

Explain the function of database object:	
Database Object	Function
Table	Tool in a database system used to store and enter data in rows and columns
Form	Tool in a database system used to enter and update record by using a formatted window screen.
Query	Tool in a database system used to request for a specific type of data by using certain criteria.
Report	Tool in a database system used to view or print data from table or query in a specific layout.

NETWORK

Describe several uses of network:	
The Internet	Largest computer network that allow the use of apps for information retrieval, online shopping, online entertainment and online communication
Telephone Service	Network communication that provide services to landline phone and cellular phone.
TV&Radio broadcasting	Telecommunication network that broadcast digital TV and radio online
GPS Application	Satellite navigation system used to determine the ground position of an object..
Monitoring System	Networking technology used to determine the current location or status of an object.
Multimedia Networking	Used of networks to deliver multimedia content to devices on the home network
Video Conferencing	Use of networking technology to conduct real-time, face-to-face meetings between individuals physically located in different places.
Collaborative computing	Networking technology with collaborative software tools to enable individuals to work together on documents and other project components
Telecommuting	Individuals work from home and communicate with their place of business and clients via networking technologies
Telemedicine	Use of networking technology to provide medical information and services
Telesurgery	Robot-assisted surgery where a robot controlled by a physician remotely do operations on the patient.

Describe wired and wireless networking media:	
Wired	Wireless
Used of physical media such as cable to send signal through network.	Used of wireless medium such as air, water and vacuum to send signal through network
Twisted-pair, coaxial, fiber-optic cable	Cellular radio, microwave & satellite, infrared
Suitable for short distance communication	Suitable for long distance communication
Fiber-optic cable is the most higher bandwidth and longer distance cable for sending signal	- Cellular radio transmission can penetrate building - microwave and satellite can send a long distance signal - Infrared is for a short distance transmission - Both microwave, satellite and infrared cannot penetrate obstacle

Describe networking device:	
Network adapter	device that connect computer to a network.
Modem	Device that convert digital signal to analog signal and vice versa
Wireless Access Point	Device that allow wireless communication between computer and devices in a network
Hub	Central device that send signal from sender to all the nodes without knowing the real address of the receiver
Switch	Central device that send signal from sender to the correct address of the receiver
Router	Central device that connect multiple device and send signal from sender to the correct receiver. Allow networks to be connected to the internet.
Repeater	Device that regenerate signal before it become weak or corrupted

INTERNET

Identify type of search tool:	
Search engine	Subject directory
Search tool that use keyword to find information that related to specific topic	Search tool that classifies webpages in anorganized set of categories and subcategories.
Users know what keyword to use to search for specific information	Users do not have precise idea on what to search and has to be guided by subject in categories and subcategories
Faster way to find information	Take sometimes to browse information by using category and subcategory
Eg: Google Search, Yahoo! Search	Eg: Google Directory, Yahoo! Directory

Deescribe different search startegies:	
Using Phrases	Use more than one keyword in keyword search
Using Boolean Operators	Use AND, OR, NOT and “ “ in keyword search
Using field search	Limit search to a particular characteristic such as title, web address
Importance of using Search strategy:	
- to help refine the specific and increase the accuracy of search result	

Describe type of Web activities:		
Web Activity	Description	Example
Instant Messaging	online chat allowing real-time text transmission over a network.	WhatsApp, Telegram
Text messaging	composing and sending electronic messages between two or more users of mobile devices	SMS, MMS
Twittering	Internet activity that post short updates about what you are doing or thinking at the moment on the Social Network	Twitter
Social Networking Update	online platform which people used to build social networks with other people who share similar personal interests	Facebook, Instagram
Forum	Not a real-time online discussions with a large group of individuals about specific topic	Extra terrestrial Forum, K-pop Forum
Voice over Internet Protocol (VoIP)	Internet telephony that allow voice call over the Internet	WhatsApp call, Telegram call
Web conference/ Webinar	used for meeting or training between individuals or group located in different location.	Skype, Google Hangout
e-Commerce	activity of electronically buying or selling product or services online or transmitting fund or data over the Internet	Online banking, online investing, online shopping, online auction
e-Entertainment	Downloading, playing and enjoying entertainment media such as games, music and videos over the Internet	Spotify, YouTube, Netflix, Nintendo
e-Education	the process of education that takes place over the Internet	Google Classroom, Google Meet, Kahoot, Quiziz
e-News & Reference	TV station, newspaper and magazine provide updated local and world news	CNN, Al Jazeera, BBC

Describe Privacy:
Enclose what information about individuals is available , how it is used and by whom.
Web browsing Privacy
Cookies-small text files that are stored on user's hard drive by Web servers when users view their web site and webpages.
Spyware- software program that is installed without the user's knowledge that secretly gathers information about the user
Adware-software that display unwanted pop-up advertisement on computer screen
e-mail Privacy
refers to keeping e-mail secured either while it transit or while it is stored in a server
How to protect:
- Set web browser privacy settings to private browsing, controlling activity logs and deleting cookies - Use antivirus or Spyware and Adware remover to remove unwanted programs that can interrupt the computer system. - Don't use personal e-mail platform for business - Always enable 2 steps verification - Do not reply spam e-mail

Discuss latest issues of Internet:

The Fourth Industrial Revolution (4IR)	Internet of Thing (IoT)
The Fourth Industrial Revolution (4IR) conceptualizes rapid change to technology, industries, and societal patterns and processes in the 21st century due to increasing interconnectivity and smart automation.	describes physical objects (device) with sensors, processing ability, software and other technologies that connect and exchange data with other devices and systems over the Internet or other communications networks.
4IR is changing how we live, work and communicate through the use of technologies like: ◇ artificial intelligence (AI) ◇ autonomous vehicles (AV) ◇ Internet of Things (IoT)	These smart devices will communicate with each other to provide advantages such as • making our lives more convenient, • saving us money, and • providing us with better healthcare and other services.
Example of apps: Chatbot- use to answer customer queries in e-Commerce platform Google AI-Powered prediction – use to analyse the speed of traffic movement and report construction and accident in Google Map Facebook Face Recognition- allow the apps to highlight face and identify friends Alexa- AI-powered personal assistant that accepts voice commands	Example of IoT: Smart heating and Cooling System to adjust temperature Smart appliances such as robot vacuum, drone, 4D printer Uses a sports-tested accelerometer to track your daily activity
Advantage	Disadvantage
making our lives more convenient	Job redeployment because work is replaced by robot
saving us money and energy	Less social skill and face to face communication
providing us with better healthcare and well being	Irresponsible used and manipulate of providing us with better healthcare technology